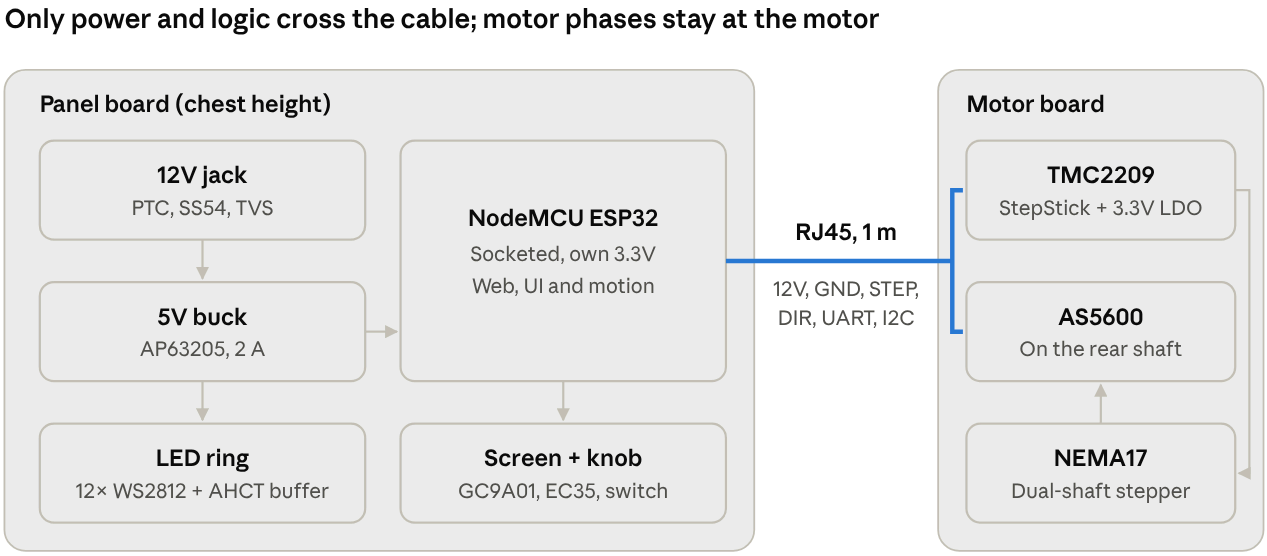
Curtain control panel: hardware design

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The panel board carries the ESP32, power, LED ring, screen and knob. The TMC2209 moves to the motor board so only DC power and logic travel through the 1 m RJ45 cable. The button is the screen itself, floating on a tact switch on the knob axis.

At a glance



system overview · panel board, cable, motor board

The panel takes 12V, makes 5V for the ESP32 and LEDs, and sends 12V plus logic down the cable; the motor board drives the coils and reads the shaft angle.

Power

One fixed-output 5V buck feeds everything on the panel; the NodeMCU's own regulator makes 3.3V, so no separate LDO is needed.

Stage	Part	Why
Input	12V barrel jack	12V 2A (24 W) adapter covers motor, LEDs and logic.
Fuse	PTC, 2A hold	Resets itself after a short.

Stage	Part	Why
Reverse polarity	SS54 Schottky in series	About 0.5V drop, irrelevant at 12V; simpler than a P-MOSFET circuit.
Surge	SMBJ15A TVS to GND	Clamps hot-plug ringing below the TMC2209's 29V limit.
Motor feed	12V to the RJ45, 100 µF / 25V + 100 nF at the driver	The bulk cap sits next to the TMC2209 on the motor board.
5V rail	AP63205 buck (fixed 5V, 2A, 32V max input)	Few external parts and no feedback divider to get wrong. Copy the datasheet layout exactly.
5V loads	NodeMCU 5V/VIN pin, WS2812 ring, 74AHCT1G125	
3.3V rail	NodeMCU's onboard regulator, from its 3V3 pin	Small loads only: LCD and knob pull-ups.

- **Avoid 17V-max bucks** on a 12V input: hot-plugging a barrel jack into ceramic caps can ring toward twice the input voltage.
- **Derate ceramic caps.** A 10 µF 25V 0805 cap may be about 4 µF at 12V. Use 1206 parts rated 50V, or two in parallel.
- **USB while 12V is on.** Most 38-pin boards have a diode between USB 5V and the 5V pin. Check yours before plugging USB with the adapter connected.

LED ring (WS2812)

Run the LEDs from 5V and drive their data line through a 5V buffer. At 5V a WS2812B wants about 3.5V on its data input, so the ESP32's 3.3V is out of spec: it often works on the bench, then flickers in the field.

- **Level shifter.** 74AHCT1G125 powered from 5V, OE tied to GND. It accepts 3.3V in and outputs 5V.
- **Series resistor.** 33–100 Ω after the buffer, close to the first LED.
- **Pull-down.** 10k on the ESP side, so the LEDs stay dark during boot.
- **Decoupling.** 100 nF per LED, plus one 220–470 µF bulk cap where 5V enters the ring.
- **Current budget.** About 1 A at 5V. Cap brightness in firmware, for example with FastLED's `setMaxPowerInVoltsAndMilliamps`.

- **Look.** SmartKnob View uses 8 side-firing SK6812-SIDE-A LEDs and white soldermask so the board reflects the light (SmartKnob). Pair that with a diffuser ring printed in white or translucent PETG under the knob skirt.
- **Layout.** 12 LEDs at r 24 mm, outside the encoder body (see PCB layout).

Pin map (38-pin NodeMCU, WROOM-32)

Your `config.h` pins all stay except motor EN, which leaves the panel because the driver moves to the motor board. That frees a safe pin for the LEDs.

Function	GPIO	Goes to	Notes
I2C SDA / SCL	21 / 22	RJ45 → AS5600	Lower `I2C_FREQ_HZ` to 100 kHz; add 47–100 Ω series resistors toward the cable.
Motor STEP / DIR	27 / 14	RJ45 → TMC2209	GPIO14 toggles briefly at boot; harmless while the driver is disabled.
TMC UART TX / RX	17 / 16	RJ45 → PDN_UART	TX through 1k joins RX on the panel; one wire goes down the cable.
Motor EN	none		Tied low on the motor board; disable over UART instead.
Knob A / B	32 / 33	EC35 encoder	10k pull-up, 10k series, 10 nF to GND.
Knob switch	13	Centre tact switch	10k pull-up + 100 nF.
TFT SCLK / MOSI	18 / 23	GC9A01	Native SPI pins.
TFT CS / DC / RST / BL	5 / 19 / 4 / 25	GC9A01	CS idles high, the level GPIO5 needs at boot. Backlight through a MOSFET if the module has none.
WS2812 data	26	74AHCT1G125	10k pull-down.
12V sense	36	100k / 22k divider	ADC1, works with WiFi on.
Ambient light	39	Phototransistor	ADC1, works with WiFi on.
Spare inputs	34 / 35		Input-only; external pull-ups if used.

Leave unused: GPIO 0, 2, 15 (strapping), 1/3 (USB serial), 6–11 (flash) and 12 (strapping: a pull-up on it stops the ESP32 booting).

Knob detents. The EC35 is 15 pulses / 30 detents, which is 2 transitions per detent, so `KNOB_TRANSITIONS_PER_DETENT 2` is already right.

Firmware changes

1. `I2C_FREQ_HZ` 400000 → 100000.
2. Remove `PIN_MOTOR_EN`; enable and disable the driver over UART (`toff(0)` to disable, your normal `toff` value to enable).
3. Add the LED pin (GPIO26), 12V sense (GPIO36) and light sensor (GPIO39).

If you keep the driver on the panel instead (option A): move EN from 12 to 26 with a 10k pull-up, put WS2812 data on 12 with a 10k pull-down, and wire TMC DIAG to 34.

Motor link (RJ45, 1 m)

Put the TMC2209 on the motor board (option B). At 1 m, I2C capacitance is no concern, but sending the motor phases through the cable (option A) has three problems.

1. **Current per wire.** Each phase would put 1–1.4 A on one 24AWG conductor. Guidance for 24AWG Cat5e is about 0.3–0.58 A per conductor (O'Reilly, PoE; RJ45 current discussion), and PCB jacks are typically rated 1.2–1.5 A per contact.
2. **Noise.** Chopped 12V phases with fast edges would run beside I2C, which was never meant for cables.
3. **Hot-unplugging.** Unplugging the RJ45 with the motor energized can kill the TMC2209 through the coils' inductive kick.

With the driver at the motor, only DC power and logic cross the cable, the supply current is lower than the coil current, and unplugging can't hurt the driver. The motor board gains a TMC2209 (a StepStick on headers is fine), a small 3.3V LDO from 12V for VIO and the AS5600, and the 100 µF bulk cap.

Option B pinout (T568B straight-through cable; pins 3 and 6 are a pair, not 3 and 4)

Pair	Pins	Standard (supply ⩽ 0.7 A)	High current (supply > 0.7 A)
1	1–2	12V / GND	12V / GND
2	3–6	STEP / GND	12V / GND

Pair	Pins	Standard (supply ⩽ 0.7 A)	High current (supply > 0.7 A)
3	4–5	SDA / DIR	STEP / UART
4	7–8	SCL / UART	SDA / SCL

- **Pick the column after a current test.** Set run current over UART to the lowest value that moves the curtain reliably, plus about 30%, then measure the 12V supply current. Curtains usually need far less than a 3D printer axis.
- **High-current column.** It drops the DIR wire: set direction over UART with the TMC2209's `shaft` bit. One small firmware change.
- **EN.** Tied low on the motor board. Put a divider on VREF so the default current at power-up, before the ESP32 configures the driver, is low.

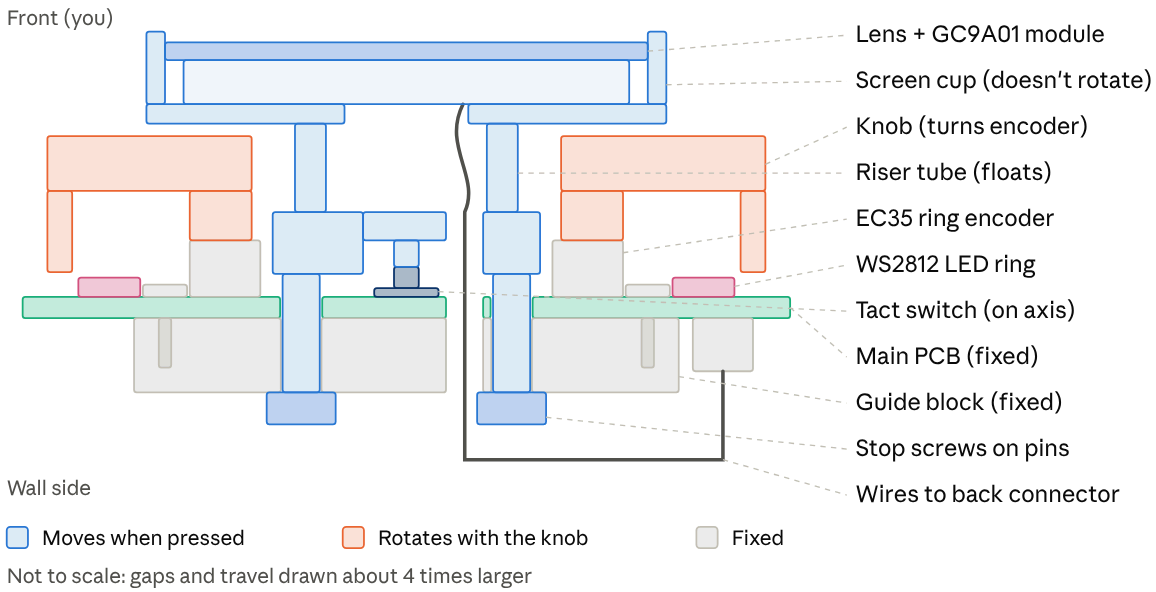
Cable and jack rules

- Plain 8P8C jacks **without magnetics**. A MagJack's transformers block DC and I2C.
- Pure-copper 24AWG cable. Cheap patch cords are often 26–28AWG or copper-clad aluminium.
- Label the port "NOT ETHERNET": 12V into a router damages it.
- **AS5600 at 3.3V:** tie its VDD5V and VDD3V3 pins together, and keep its DIR pin tied to GND as in your README.
- The magnet on the rear shaft needs a **dual-shaft** NEMA17.
- **Test before ordering.** Run your dev boards through the actual 1 m cable and log AS5600 read errors while the motor turns.

If you keep option A anyway: shielded Cat5e/6 with shielded jacks, shield grounded at the panel end; I2C at 100 kHz or less; motor current at or below about 0.8 A RMS; never unplug while powered. Pinout: 1–2 coil A, 3–6 coil B, 4–5 SDA / GND, 7–8 SCL / 3.3V.

Button: press the screen

The knob only rotates the encoder; pressing the screen moves a separate riser 0.4 mm onto a tact switch on the knob axis. The PCB, encoder and connectors never move.



control panel cross-section · center-press mechanism

The riser, screen and stop screws move together; the switch clicks first, then the foot lands on the PCB.

The push function has to be your own mechanism: ALPS ring encoders such as the EC35A have no push switch, and the EC35B's switch is a spring-return twist of about 48°, not a press (EC35A, EC35B). The EC35 tolerates 100 N of push/pull, so loads on the knob are no concern.

Option	Verdict
Press the screen on a floating riser	Chosen. One small part moves; long guide; tunable with shims.
Whole front floats on 3 guided posts (Nest-style)	Works, but rattle versus binding takes iterations, and the jack and cable move with it.
PCB on springs and rubber pads	Wobbly and inconsistent; tilts when pressed off-centre.
PCB flexure cut-outs	SmartKnob senses micro-deflection with strain gauges, not a click; a tact click needs tuned FR4 arms with fatigue risk (SmartKnob build notes).

Option	Verdict
Round LCD with capacitive touch (CST816S)	No mechanics, but no click.

Parts

1. **Riser.** One printed piece: a screen cup, a tube through the encoder's hollow centre (it never touches the rotor), and a foot disc about 21.6 mm across with a nub on the switch and a window for the screen wires.
2. **Guide pins.** Three M3 screws in heat-set inserts in the foot. They pass through clearance holes in the PCB and slide in about 6 mm long bores in a printed guide block on the back of the PCB. The PCB's 1.6 mm alone would rock.
3. **Return spring.** The tact switch pushes the riser back; the screw heads hitting the back of the guide block set the rest position.
4. **Hard stop.** The switch clicks at about 0.25 mm; the foot lands on the PCB at 0.4 mm, so a hard press loads the PCB, not the switch.

Tuning without reprinting. Add 0.2–0.5 mm shim washers under the stop-screw heads. Each shim pulls the riser closer to the switch, adding preload and an earlier click.

Anti-rotation. The three pins stop the riser turning, so the screen wires can't twist. Leave about 10 mm of slack; they only move 0.4 mm.

Assembly order

1. Solder the PCB, including the encoder and switch.
2. Push the riser foot in from the front through the encoder bore.
3. Screw the guide block on from the back.
4. Insert the three screws from the back into the foot inserts.
5. Clip the knob onto the encoder rotor, keeping at least 1 mm of axial gap below the cup.
6. Mount the screen in the cup.

Practical tips

- **Switch.** A 5.2 × 5.2 mm SMD tact switch (TS-1187A type) fits beside the wire slot and JLC can place it. Any 160–260 gf switch works.
- **Printing.** PETG for the riser and guide block. Print the guide bores vertically and ream them with a 3 mm drill.

- **Test before the PCB order** on a scrap of 1.6 mm board, or a printed plate, with the same holes.

PCB layout around the encoder

Inside the encoder bore the PCB holds only the switch, a wire slot and three pin holes. Dimensions assume the EC35A's 23.6 mm bore (EC35A datasheet); check them against your encoder's drawing before laying out the footprint.

Feature	Size	Position (from knob axis)	Side
Tact switch	5.2 × 5.2 mm SMD	Centre	Top
Guide pin holes	3× Ø3.4 mm NPTH	r 8.5 mm, at 0°, 120°, 240°	Through
Wire slot	3 × 6 mm, Edge.Cuts	r 3.2–6.2 mm, at 180°	Through
Riser foot outline	Ø21.6 mm	Centred	Top (keep clear)
EC35 encoder body	Ø35 mm, bore Ø23.6 mm	Centred	Top
Guide block screws	3× M2	r 19.5 mm, at 90°, 210°, 330°	Through
WS2812 ring	12 LEDs, 5 × 5 mm	r 24 mm, every 30°	Top
Guide block keep-out	about Ø44 × 7 mm	Centred	Back
Screen wire connector	8-pin JST XH	Beside the guide block	Back

Nothing else goes inside the bore. The NodeMCU socket, buck and RJ45 sit outside the guide block keep-out on the back.

Fixes in the current KiCad schematic

The schematic in `hardware/blinds-pcb` has its symbols placed but nothing wired yet, so these are cheap to fix now.

- **RJ45 footprint.** `RJ45_HALO_HFJ11-x2450HRL` is a MagJack footprint (built-in Ethernet transformers). Replace it with a plain 8P8C jack without magnetics and the footprint for that exact part.
- **Driver symbol.** The Pololu DRV8825 symbol doesn't match a TMC2209 StepStick. Its FAULT pin is VIO (3.3V logic supply) on a TMC2209, and its RST/SLP positions are PDN/UART pins whose layout varies by vendor. Draw a symbol from your module's silkscreen; with VIO unconnected the driver won't work. With option B this symbol moves to the motor board's schematic.
- **Missing footprints.** Ten symbols have no footprint assigned yet (switch, encoder, capacitor, resistor and others).
- **Missing parts.** Add the power stage, LED level shifter, encoder RC filters and the 12V sense divider from this doc.

Board, ordering and bring-up

Design the board so each block can be powered and tested on its own; that is what makes a first board work.

Layout and ordering

- **4 layers.** Cheap at JLC, and a solid ground plane makes the buck and ESP32 layout forgiving.
- **Antenna.** The NodeMCU's antenna overhangs a board edge or sits over a copper keep-out, away from the LCD and the encoder's metal body.
- **Parts.** Prefer JLC basic parts (each extended part adds a fee). Hand-solder or THT-assemble the jack, RJ45, encoder and sockets.
- **Export.** The KiCad "Fabrication Toolkit" plugin exports gerbers, BOM and CPL with LCSC part numbers; check part rotations in JLC's preview.
- **Traces.** 1 mm for 12V input; 0.5 mm or more for 5V to the LED ring.
- **Modules.** Socket the NodeMCU, and use a TMC2209 StepStick on headers rather than the bare QFN: no thermal-pad layout to get wrong, and a dead driver is a swap.

Bring-up

- [ ] Solder jumpers or 0 Ω links between each regulator and its loads; test points on 12V, 5V and 3.3V; LEDs on 5V and 3.3V.
- [ ] First power-up from a current-limited bench supply with the loads disconnected; measure the rails.
- [ ] Close the jumpers one at a time: NodeMCU, LCD, LED ring, then the motor board.
- [ ] Run your existing test firmwares (`knob` , `display` , `encoder` , `motor` , `closed_loop`) in that order.

Extra features worth adding

- **Stall and obstruction detection.** Read StallGuard's `SG_RESULT` over UART to stop if the curtain snags on something, and for sensorless homing. (DIAG has no wire in the option B cable.)
- **Ambient light sensor.** Auto-dim the screen and LEDs at night; SmartKnob uses a VEML7700 for this (SmartKnob). A phototransistor on GPIO39 is the simplest version.
- **12V sense.** Detect power loss and save the position before the rails collapse; the AS5600 only knows its angle within one turn.
- **Pseudo-haptics.** A small piezo or coin vibration motor through a transistor can tick on each detent.
- **Expansion.** A spare 4-pin I2C (Qwiic) header, plus BOOT and RESET reachable without opening the case.

Open questions

- [ ] Is the measured 1–1.4 A the 12V supply current or the coil current? It decides the cable pinout column.
- [ ] Exact EC35 part number, to confirm the bore and footprint dimensions.
- [ ] Does the GC9A01 module have its own backlight transistor?
- [ ] Does the NodeMCU have a diode between USB 5V and the 5V pin?

Sources

- ALPS EC35A ring type datasheet
- ALPS EC35B composite type datasheet
- Power over Ethernet, N-pair power estimates (O'Reilly)

- Current rating of an RJ45 connection (discussion)
- SmartKnob on GitHub (fork)
- SmartKnob build video transcript
- automated-curtains repository